Case Analysis: CatalanGate

In 2022, a political scandal erupted when reports detailed the hacking of prominent people from Catalonia, specifically through their use of the popular mobile/instant messaging software WhatsApp. Catalonia is currently an autonomous region within Spain, and, in conflict with the interests of the Spanish government, has sought independence throughout modern history. The people affected by the hack include elected officials, political leaders, and activists that were involved in the Catalan independence movement, as well as people associated with them, all of whom were users of WhatsApp. Much of the hacking occurred around the time of the independence referendum in 2017. The most in-depth report of the breach of Catalan citizens (Scott-Railton et al, 2022) was published by Citizen Lab, a prominent research laboratory at the University of Toronto that studies issues in cyberspace. They detail targeted hacking, alleged to be primarily organized by the Spanish government. The investigators dubbed the incident “CatalanGate,” using a popular portmanteau of the 1972 Nixon Watergate scandal, to describe what they declared to be a surveillance and hacking campaign by the government of Spain against their perceived Catalan political opponents.

The Citizen Lab report was released following a broader journalistic investigation called the Pegasus Project, a collaboration between multiple organizations, that revealed the breadth of surveillance hacks in Europe and beyond. However, even within just the CatalanGate breaches, the scandal continued in multiple venues, including the news, courts, and personal opinions worldwide. The conflict implicated a variety of stakeholders, across multiple jurisdictions, in a changing environment. The affected users were Catalan people, who are citizens of Spain, and protected by the recently applied (as of 2018) General Data Protection Regulation of the supranational European Union, and rights protected by the UN. The software producer is WhatsApp, an American company with massive reach in Europe, purchased by Facebook in 2014, and now part of the parent conglomerate Meta. Responses to the scandal include multilevel EU enforcement activities and US court decisions in 2025.

The CatalanGate scandal, as described by the Citizen Lab report, is actually an agglomeration of multiple cybersecurity incidents that occurred primarily between 2017 and 2020. This report (Scott-Railton, 2022) is an instrumental source of information about the case, and a crucial foundation for further analysis and action by many stakeholders. An earlier investigation, The Pegasus Project also shed light on the broader issue of spyware starting in 2021 (Amnesty International, 2022). Additionally, after the discovery of these incidents, the European Parliament conducted a series of

investigations which, crucially, yielded a Report created by Rapporteur Sophie in ‘t Veld, which was formally adopted by the EU (2023). I rely on these reports and investigations heavily for my case analysis, as they include data that I would not be able to replicate, such as interviews with the affected users, technical details about the breaches, and legal analysis. However, I do not need to reproduce their findings and instead can extend their analysis with additional data from before and, especially, after the incidents.

1 Cyber Social Context

The cyber social context overarches the environments in which the stakeholders of the case operate and sets the stage for the incident and its aftermath. Per the model, the stakeholders are the users, software producer, and regulators. Each of them has a regulatory, threat, and operational environment. In this case, the users were public figures involved in the Catalan government and civil society, as well as some “off-centre” victims, people that are simply related to those people, like family members, all of whom were using the mobile app WhatsApp (Scott-Railton et al, 2022). The software producer is WhatsApp, a popular mobile messaging application. Stakeholder regulators include the Government of Catalonia, the Government of Spain, the European Union, and the United States government.

1.1 Users

WhatsApp was acquired by Facebook (now Meta) in 2014. At the time, WhatsApp boasted 465 million monthly active users and Facebook itself had 1.2 billion users. By the end of 2017, WhatsApp had over 1.5 billion users and in February 2020 they reached 2 billion (Richter, 2020). For years, WhatsApp was clearly the choice for mobile messaging users worldwide, both before and after it was acquired by social media monolith Facebook. Due to its prevalence, it naturally included everyday users that were of interest to prospective political opponents that sought their private information.

In this case, general WhatsApp users fall into two crucial categories. First, they are EU data subjects, as citizens of Spain, a member state of the EU. Second, they are affiliated with the Catalan government, which has a history of conflict within Spain.

Additionally, the users that were ultimately compromised in the case also fall into the category of “high risk” targets, including journalists and independence movement activists who are often the subjects of government surveillance.

1.1.1 Operational Environment

Between 2014 and 2017, WhatsApp was one of the most popular, if not the most popular, mobile chat and messaging applications in the world, adding “roughly 25 million new active users each month, or 833,000 active users per day” (Olson, 2014). As of 2024, WhatsApp is the most popular social media platform in Spain (Statista, 2025). For everyday messaging between people, it is clear that WhatsApp was the preference for many people, including in Spain. However, as strictly free, consumer users, WhatsApp users fall far down in a hierarchy of operations with the company, and have no direct relationship with the developers.

In the Citizen Lab report, they describe the use of Android devices compared to Apple iOS devices. This is a commonplace discussion between users of chat applications, and for this case study is irrelevant. At the end of the day, WhatsApp was incredibly popular and, retrospectively, we can see that its use resulted in a cybersecurity incident.

Targeted Users

The Catalangate report concludes that 63 people were targeted with spyware, and 51 were infected and their data compromised. Of those, 47 were targeted with the Pegasus spyware that exploited a vulnerability in WhatsApp, and 18 of those were confirmed to have been conducted by the Spanish government. Confirmed infections include Members of European Parliament (MEPs), civil society figures, Members of Catalan Parliament, and all Catalan Presidents since 2010, as well as individuals who had personal relationships with those people (Scott-Railton et al, 2022; Veld, Report, 2023).

Outside of Spain and Catalunya, there were documented cases of more than 1400 WhatsApp users being targeted with the Pegasus spyware (Amnesty, 2025).

Advocacy

A rather large and active network of advocacy organizations exists to support data privacy and security as they pertain to human rights, democracy, and media/journalism. Organizations like Citizen Lab, Electronic Frontier Foundation, Access Now, and Amnesty International investigate and report on cybersecurity issues, including threats to civil society and democratic institutions. They also attempt to influence regulations.

One example is a 2018 report from Citizen Lab about Pegasus, prior to the discovery of the Catalangate breaches. This report provides technical documentation about the software including IP addresses and domain names associated with the malware, as well as some of the actors using it around the world (Marczak et al, 2018). Artifacts like this one can be reported to authorities for further investigation and action.

1.1.2 Regulatory Environment

WhatsApp users in Spain, and in Europe more broadly, fell under the regime jurisdiction of the European Union's General Data Protection Regulation (GDPR) as of May 25, 2018. It has been described as a possible basis for a "gold standard" for global data privacy, and includes a number of protections for users that require action by businesses, and mandates for governments (Rustad & Koenig, 2019). In particular, it protects the personal data of EU consumers, based on the premise that "all natural persons have a fundamental right to the protection of personal data" including right to be forgotten, right to object, right to rectification, right of portability, right of access, and right to be notified (Rustad & Koenig, 2019). Additionally, they give EU data subjects the right to collectively complain against violators, similar to US-based class action lawsuits. EU data subjects are also protected from "personal-data record-keeping systems whose very existence is secret." These protections apply to users of foreign businesses that offer goods or services in the EU, not just companies based there.

However, there are some restrictions to the GDPR protections. For example, the "Right to be Forgotten" is a protection that allows data subjects to have information deleted after a certain period, the right to have a "clean slate," and the right to access and clear outdated information. However, this right does not preclude legal obligations which require data processing by law, the exercise of freedom of expression, for archiving purposes, and more (Rustad & Koenig, 2019). More importantly, the GDPR protections are limited in matters of national security, see Section 1.2.1 for more details.

1.1.3 Threat Environment

WhatsApp & Technical Threats

Due to the scale of its user base, WhatsApp is a prized target for cyberattacks. In a 2023 incident, the phone numbers of WhatsApp users were allegedly made available for sale, including nearly 11 million people in Spain (Lapienyte, Cybernews, 2023).

By 2019, the vulnerabilities exploited in the Catalangate scandal were known to WhatsApp and its then-parent company Facebook. In this case, the vulnerability was known to a company called NSO Group, who sold the exploit to customers for profit. Roughly 1,400 users were targeted through the WhatsApp exploit known as CVE-2019-3568, including 47 in the Catalangate scandal (Scott-Railton et al, 2022). WhatsApp and Facebook sued NSO Group in US Federal Court in 2019 (Penney & Schneier, 2022), ultimately leading to a win for Meta in 2025.

Political Threat Surface & Surveillance

According to Citizen Lab, there have been multiple recent cases of possibly illegal surveillance activities by Spanish authorities, including purchases of spyware between 2001 and 2020. In particular, a 2017 referendum on Catalan independence seems to have been the impetus for the data breaches between 2017 and 2020 (Railton et al, 2022). The users targeted in this case (see Section 1.1.1) were high value/high risk targets for the attackers (Interview C004).

As potential victims of surveillance for security or political reasons, these users are at increased risk. According to a vice-president at NSO in an interview with The New Yorker, in order to create products that are valuable to government customers for surveillance, they are "looking for a silver bullet, a simple exploit that can cover as much mobile devices around the world" (Farrow, 2022).

Prior examples of Spanish users targeted by spyware include the Spanish authorities authorized breaches in anti-drug efforts, combating terrorism, and fighting corruption (Veld, 2023).

1.2 Regulators

In this case, the clear regulatory stakeholders are the agencies and courts of the European Union that are responsible for data protection under the GDPR. This primarily includes the institutions of the EU, such as the European Parliament, European Commission (EC), and the Court of Justice (CJEU) headquartered in Brussels. The case also implicates the governments of Spain, where the conflict primarily took place, Ireland, where WhatsApp maintained its European headquarters at the time, and the US, where WhatsApp was primarily based. The case takes place between 2017 and 2020, which was also a period of transition for regulators in Europe with the implementation of GDPR in May 2018, and subsequent interpretations of the new law by legislators and jurists.

The Spanish government, in addition to being one of the responsible regulators for the incident, is also confirmed to be the perpetrators of at least some of the hacks, specifically the National Intelligence Service (CNI) and the Ministry of Defense. The agencies responsible for the hacks used legal procedure to justify them, both before and after the incidents, which add to the complexity of the rules that need to be analyzed.

1.2.1 Regulatory Environment

1.2.1.1 Background

The European regime, particularly under the EU, "highly prizes the protection of personal information" (Prost, 2018), and sees privacy as "a key democratic value …

[and] favours precautionary measures" to protect EU citizens' data "held or processed anywhere in the world" (O'Hara & Hall, 2021). This regime did not begin with the establishment of GDPR and is based on long legal histories from its member states. Immediately preceding GDPR was the 1995 Data Protection Directive, which was quite strong in its protection of EU citizens, and bolstered by the 2009 Treaty of Lisbon (O'Hara & Hall, 2021; Prost, 2018). However, the EU was induced to expand the protection of EU data subjects through GDPR due to key societal changes: First was the general growth of the internet over the prior two decades and the development of monolithic companies like Google and Facebook; second was a 2015 CJEU decision, known as Schrems I, against Facebook regarding the transfer of EU citizens data to the US in light of the leaks from Edward Snowden; and third was a case against Google Spain regarding the right to be forgotten (O'Hara & Hall, 2021; EPIC, 2016; Anthony, Ars Technica, 2015; Cox, Ars Technica, 2020). The net effects of these conflicts were to alter existing regulations or initiate new ones, such as GDPR and special agreements with the US, such as a data transfer agreement called Privacy Shield. However, they did not provide compensation for the complaining parties.

The series of action and enforcement in the Schrems cases demonstrates the prioritization of user protections by the EU and its institutions. Additionally, it connects Spain specifically as a member state that participated directly in the emergent regime governing data privacy and security. However, they are *not* examples of precedence for collective action against companies for data breaches.

1.2.1.2 Surveillance and Spyware in Europe

This case includes the use of surveillance by national security regimes, which is also regulated in the European Union, including by the CJEU and the European Court of Human Rights (ECtHR) (European Parliament, Investigation of the Use of Pegasus, p9_TA(2023)0244, 2023). The interests of national security and individual rights need to be balanced in cases like this, and use of spyware for "political purposes, targeting critics and opponents of the parties in power" can be considered abuse of the rights to use spyware by states (Veld, Report, 2023).

1.2.1.3 Spain

Although Spain is a Member State of the EU, it also maintains its sovereignty and has its own regimes in place. Even the authority of the EU can be precluded from interference in certain situations, specifically on national security (Veld, Report, 2023).

The 2023 Report on the Catalangate and broader Pegasus, written by Sophie in 't Veld, and adopted by the European Parliament provides useful information about the structure of the Spanish regime on user protections for data privacy. The report explains

> The right to privacy is protected under Article 18 of the Spanish Constitution of 1978, including the right to secrecy of communication, and in particular guarantees 'postal, telegraphic and telephone communications' … The constitution also provides for further exceptions to those rights in Part I Section 55 by stating that some rights can be suspended subject to the 'participation of the courts and proper parliamentary control' if it has been agreed to declare a state of emergency or siege under the terms provided for in the Constitution or in the case of individuals under investigation for activities relating to armed groups or terrorist organisations. Furthermore, Article 55 includes democratic safeguards to ensure that 'unjustified use or misuse' of those powers will give rise to criminal liability.

In short, the privacy of Spanish citizens is protected, but the emergent regime may be altered by state security interests if they need to violate those protections. There are a number of legislative and judicial decisions that inform the ability of the government to suspend those protections, including both Spanish and European laws. They require specific authorization and oversight by various government bodies. Broadly, the authorization to suspend protections are based on four principles: "firstly, specialisation (that the surveillance is related to a specific crime); secondly, adequacy (outlining duration, objective and the subjective scope); thirdly, proportionality (strength of available evidence, severity of the case and result sought), and finally exceptional nature and necessity (there are no other measures available and without it, the investigation will be interfered with)" (Veld, 2023; Maciejewski et al, 2023). It is worth noting that the Spanish Data Protection Agency has no power over the National Intelligence Service (CNI) (Gortazar et al, 2016).

1.2.1.4 Other International Regulators

As a regulatory stakeholder in this case, the UN's primary contribution to the regime in this case is related to human rights, such as through the Office of the High Commissioner on Human Rights (OHCHR). Other international agreements can also have an impact on the context. Prior to the Catalangate case, there were few salient examples of international regulation by the UN or other regimes regarding spyware.

1.2.2 Operational Environment

The EU and its institutions were empowered to protect user privacy and data security, as were its member states. However, it is possible that the interests of member states like Spain were at odds with EU laws, particularly in the realm of surveillance.

The European Parliament has the authority to conduct investigations into matters of data security and privacy with regard to its member states and their citizens. Despite the landscape of regulatory enforcement against major tech companies, the EU's approach to spyware has been described as a "wild west of spyware" as it lacks rules on licensing, export laws, who can use it, and how it is used (Roussi, 2023). It seems as though there is a conundrum, where EU lawmakers like to have enforcement methods against big tech companies, especially those based in the US, but are reluctant to regulate spyware, where their own national security interests may be at risk.

In Spain, the primary operational context for regulators in this case deals mainly with the Catalan independence movement, against which the Spanish government has aggressively fought. There was a tense political environment, and economic concerns regarding the possibility of secession, including the arrest of a former president of Catalonia, and sometimes violent repression (Henley, 2017).

In general, there was a challenging hierarchy between potential targets of spyware and the regulators responsible for the regime. Regulators in this case were not closely tied to the targeted users by either the Spanish or European regimes.

1.2.3 Threat Environment

Broad regulatory attention to the threat of spyware arose in 2021 in an investigation called The Pegasus Project, led by journalists and civil society organizations like Amnesty International (Amnesty International, 2022). This investigation largely highlighted regulators' limited familiarity with the threat environment and implored them to act. As will be described later, the regime also protected the use of spyware threats by government actors for national security purposes.

1.3 Software Producer

As described in Section 1.1, WhatsApp was, and continues to be, the most popular mobile/instant messaging application in the world. Its growth, as measured by users, was insurmountable by competitors, which is likely what provoked Facebook to buy it in 2014 for $19 billion, reaching 2 billion users in 2020 (Richter, 2020). Facebook, at that

point, was already in the crosshairs of EU regulators (Schrems, 2015; High Court Ireland, C-362/14, 2015), and this acquisition could implicate WhatsApp in any future interest or action against Facebook. Ultimately, it meant that any regulatory engagement with WhatsApp would necessarily incriminate Facebook.

1.3.1 Regulatory Environment

As discussed in Section 1.2, software producer Facebook, Inc was the subject of consequential regulatory action in Europe. Founded in 2004, by 2014 the company had become a monolithic operator in cyberspace, controlling a substantial amount of internet users on mobile and desktop devices worldwide. In 2015 and 2020, they lost two cases in the CJEU (both of which were initiated by privacy activist Max Schrems) that were instrumental in setting precedence and establishing regulations for software producers that had users protected by the EU. The 2015 Schrems I decision was instrumental in driving the passing of GDPR, and Schrems II in 2020 instigated important interpretations of the new regulations (O'Hara & Hall, 2021; EPIC, 2016; Anthony, Ars Technica, 2015; Cox, Ars Technica, 2020; Prost, 2018; Kuner, 2017; Tzanou, 2020).

In short, Facebook was not simply the subject of regulatory action, they were an archetype of software producers upon which European regulators established their laws.

1.3.2 Operational Environment

As described above, WhatsApp was huge at the time of the data breaches. It was continuously growing, regularly adding new features and security updates to maintain its commercial viability (Angela, Online-Convert.com, 2017; WhatsApp, 2016; Koum & Acton, 2016). While Facebook and its subsidiaries like WhatsApp provide a variety of features for their users, they are not known for working closely with them, as they provide free services that are funded primarily by advertising. Unlike software as a service platforms where users contract directly with the company, WhatsApp users would be less likely to cooperate directly with the company itself.

1.3.3 Threat Environment

Messaging apps are technically challenging for software producers to secure, according to the Electronic Frontier Foundation (EFF). In 2018 they created four publications about the variable threat surface, concluding that "there is no single correct way to balance security features, usability, and the countless other variables that go into making a high-quality, secure communications tool" (Cardozo et al, 2018). For example, they mention that more popular messaging software can be more secure because it is

less suspicious (compared to using software designed specifically for secure messaging that could readily be identified for cyber attacks), and larger companies have more resources (i.e. internal familiarity) to maintain the application in the face of a complex threat environment (Portnoy, 2018). With that in mind, the EFF staff that created the publications said that they would “reluctantly” recommend average users use WhatsApp for these reasons (Gebhart, 2018). On the flip side, and adding to the complexity, an earlier 2017 EFF report about government access to user data gave WhatsApp a 2-star rating out of 5, indicating that they may not adequately protect users from government intrusion (Cardozo et al, 2017). This rating includes assessments of both their interpretation of legal regimes and internal regulations about protecting data.

In general, the scale of the WhatsApp userbase may have made it a more secure option for the average user, but it created a huge threat surface for surveillance and spyware activity by security agencies.

2 Operating & Precursors

2.1 Users

The users targeted included civil society representatives, lawyers, journalists, and elected officials that were directly or indirectly part of the pro-Catalan independence movement. They also include people who were simply affiliated with those figures, described as “by-catch” and “relational or ‘off-centre’” targets (possibly even people mistakenly targeted because they shared the same name) (Scott-Railton et al, 2022; Veld, 2023). Due to the ubiquitous use of WhatsApp, they were presumably using the app for both their personal and official/business purposes, creating potentially valuable data to attackers in either case, and providing a vector for nefarious use of their mobile phones for surveillance. The users were targeted across a long timeline, but in particular before and after key events like the 2017 independence referendum, and other political events such as elections and political appointments (Veld, 2023; Scott-Railton et al, 2022).

2.1.1 The Surveilled and their Software

The targeted users in this case were using the software WhatsApp to communicate with other people on their mobile devices, just like any other user. However, these users were seen as significant, and their use of the app created data that was desirable to would-be attackers. We do not know whether or how much of that data was actually

valuable to an attacker; it is possible that users secured critical information with hygiene practices, like discussing sensitive information on more protected platforms such as encrypted email or other internal messaging systems. For example, two of the targets were software developers that worked on security-focused technology, and their devices were not compromised. Furthermore, the breaches they suffered could have offered up more than just text messages within the app: the spyware is likely to have exposed them to additional surveillance by compromising the sensors on their mobile phones, like cameras, microphones, and GPS antennas.

Even with operations that include rigorous prevention practices, and strict adherence to security rules, these users were still at serious risk because of the level of interest by the attackers, who could use complex methods like zero-click attacks that target zero-day vulnerabilities (Scott-Railton, 2022; Khandelwal, 2019; Sartor & Loreggia, 2022; Marzocchi & Mazzini, 2022). The targeted users worked for a variety of organizations and even government agencies, and those organizations may have had adept defense and prevention security practices. They could have been secured through their IT departments across the various software that they used for work. However, for individuals targeted for surveillance, a more powerful attack outside of those applications, one focused on unaddressed failures not supported by their employers, was the larger threat, and the vector was the nearly ubiquitous messaging app WhatsApp.

People that are considered high risk for surveillance and spyware technology, like those in this case, are able to receive training and technical support from advocacy organizations like Citizen Lab, Amnesty International, and Access Now. However, those resources are limited and are by no means a guarantee of safety (Interview C004) and we do not know whether any of the targeted users took advantage of these opportunities.

2.1.2 Precipitating Changes

In many instances, the attacks occurred around the time of important events for the users involved. Basically, in the course of their operations, there were changes in their business that made their data valuable to potential attackers. The most prominent change was the 1 October 2017 referendum on Catalan independence, which had an enormous impact on Spanish and global society. The referendum passed, with 92% voting yes, but was declared unconstitutional by Spain. It also resulted in later events that were of interest to hackers like the Spanish intelligence agencies. Because

WhatsApp was already deployed on their phones, they could be targeted nearly at-will by hackers with the right resource, the Pegasus spyware.

The October 2017 referendum was a sea change in Spain, with potential impacts on the fundamental nature of the country, including its economy, politics, geography, population, institutions, etc. It should come as no surprise that the Spanish government were acutely interested in the movement, and their intelligence agencies would increase their focus on the key actors. One such actor was Jordi Sanchez, the president of the pro-independence Catalan National Assembly (ANC). Among the known breaches Sanchez was one of the earliest, and was hacked multiple times between May and October 2017 (Scott-Railton et al, 2022).

Immediately following the referendum, multiple activists promoting independence were imprisoned for their involvement, and some were freed in 2021, after over three years in prison. During their imprisonment, their legal teams and colleagues were attacked via Pegasus at critical times of change, including nearing their court cases and during discussions of how to fill a parliamentary seat left open, after it was decided one politician could not serve while imprisoned (Veld, 2023; Scott-Railton et al, 2022).

2.2 Regulators

While European regulators were working on implementing GDPR to protect data privacy and security, they did not directly address spyware. Meanwhile, Spanish authorities were pursuing court orders to legally hack phones of targeted individuals.

2.2.1 European Regime

There are overlaps between the laws of the EU and the laws of its Member States when it comes to national security, intelligence, and law enforcement. As such, the regime that governs the actions of Spain using spyware on its citizens in this case, such as its rules, enforcement, and capacity, is very ambiguous. According to the PEGA Committee Report, “all Member States use spyware in one way or another, some legitimate, some illegitimate” and at least 14 EU countries purchased the Pegasus spyware (Veld, 2023). According to the Treaty of the European Union, Member States have exclusive rights over their national security. However, the CJEU and the ECtHR also have case law that attempts to balance fundamental rights with national security considerations. In short “the lack of clarity as to how national security is defined and the excessively broad interpretation of its scope by the national authorities pose a challenge in understanding the justification for the use of spyware,” (Veld, 2023). This indicates ambiguity for the capacity of regulators, and the rights of users and software producers.

2.2.2 The Surveillance Regime: Spanish Authorities

In this section, I will investigate the regulatory operations of the Spanish authorities. What makes this case, and this section, interesting is that Spanish authorities are also the attackers that executed the breaches. However, I will not be scrutinizing the full anatomy of the attacks even though they were carried out by partners of the regulatory bodies. The focus will be on how the Spanish government implemented the regime, not the attacks themselves.

The CNI is an agency created by parliament in 2002 as a part of the Ministry of Defense. It "achieves its missions by collecting information in Spain and overseas and acts under the supervision and control of the executive, legislative and judicial powers" (Veld, 2023). Additionally, the CNI must be authorized by the Spanish Judiciary in advance, such as through a court order, "to adopt measures that may affect the inviolability of the home, and the secrecy of communications" (Gortazar et al, 2016; Veld, 2023). Furthermore, the Spanish Ombudsman also has full access to documents to investigate actions by the CNI after they have occurred, but has limited enforcement powers (Veld, 2023; Gortazar et al, 2016).

As described in Section 1.2.1, privacy rights for Spanish citizens are protected by the constitution, but there are exemptions to those rights. These exemptions are within the capacity of the CNI to exploit, given their authority, funding, and the rules made to govern their actions. In particular, the CNI is tasked with preventing threats to the "independence and integrity of the state," (Veld, 2023) which could easily be interpreted to encompass the Catalan independence movement. The CNI and other authorities implemented their regime to be able to conduct spyware operations where they saw fit, such as by getting judicial authorization in advance (Tomás, 2022).

2.3 Software Producer

When WhatsApp was acquired by Facebook in 2014 for $19 billion, it became part of one of the largest software companies in the world. By 2020, it claimed over 2 billion monthly users (Richter, 2020; WhatsApp, 2020). However, its popularity with users meant it also became a high value target for hackers.

WhatsApp's drastic business changes included user growth and acquisition, as well as updated features, which would mean expanding capacity to maintain and secure the software. They implemented prominent new features including file uploads (Angela, Online-Convert.com, 2017), voice and video calling (WhatsApp, 2016), and end-to-end

encryption (Koum & Acton, 2016). These features met the needs of their users by adding more communication and security capabilities, and were completed by a “small engineering organization” of about 57 people (Reed, 2016). This small organization approach also entailed a small technical footprint which helped them “reduce the number of things that could break” meaning that they had adequate capacity despite their minimal size (Reed, 2016). However, some of these features could also increase the threat surface for users, requiring additional security measures.

WhatsApp took data privacy and security seriously, even when they were criticized for partnering with Facebook, which itself had been criticized for privacy concerns. Co-founder Jan Koum detailed his commitment to “private communication” due to his upbringing in the USSR, asserting that “respect for your privacy is coded into our DNA” and that partnering with Facebook would not “compromise the vision” (Koum, WhatsApp, 2014). WhatsApp did undertake a multi-year “effort to reinforce the security of its products” and was able to coordinate with Facebook security capacity as well (Farrow, 2022). Security patches prior to 2020 included at least 13 patches for known vulnerabilities from the NIST National Vulnerability Database (WhatsApp Security Advisories Archive, accessed 2025), mostly from 2018 and 2019. In 2020, they patched an additional 15 known vulnerabilities (WhatsApp Security Advisories 2020 Updates, accessed 2025). Included in those patches was CVE-2019-3568, which was the unaddressed failure exploited by the Pegasus spyware, but not before it was exploited.

In its growth, and in the changing regulatory landscape, WhatsApp complied with regulatory requirements when they were defined. The most prominent example is the introduction of GDPR, which forced WhatsApp to create new Terms of Service and Privacy Policy documentation, as well as a business entity in Europe in 2018. The new business, WhatsApp Ireland Limited only serves users in the European Economic Area and the European Union, and is the data controller responsible for their services, per EU law (WhatsApp, Updating Our Terms, 2018; WhatsApp Privacy Policy, 2018).

3 Incident & Impacts

Between 2017 and 2020, around 1400 WhatsApp users in 45 countries were targeted for the vulnerability eventually identified as CVE-2019-3568, which was compromised using the Pegasus spyware technology. Of those, 47 were Catalan people directly or indirectly connected to the independence movement and alleged to be infected with the spyware (Scott-Railton et al, 2022), 18 of whom were confirmed by the attackers themselves, the Spanish National Intelligence Service (Veld, Report, 2023).

3.1 Users

The users, Catalan civil society and political figures discussed in Section 2.1.1, suffered unauthorized access of their data, including turning their devices into sensors which could be used for surreptitious surveillance. The scale of the breach for each user included text messages, emails, WhatsApp messages, contact details, call records, location, microphone and camera (Khandelwal, 2019). It is unclear exactly what this data would be used for by the attacker, and any impacts beyond the violation of their private information. As reported by many investigators, the Spanish authorities were behind the attacks and likely used it to stymy or prevent Catalan independence efforts, including arresting multiple people associated with the cause. Some of the attacks coincided with court cases, and thus the compromised data could have been used to inform legal strategy, although that cannot be verified.

Investigators describe the impacts of the Pegasus spyware as human rights violations "including monitoring, blackmailing, smear campaigns, intimidation and harassment" with secondary impacts on "data protection and privacy, freedom of expression, freedom of the press, freedom of association, redress mechanisms, legal remedy and fair trial, and democratic processes and institutions" (Veld, 2023).

3.2 Regulators

Regulator interest began in earnest following this breach, although the software behind it was known previously. The PEGA Committee would later adopt the following definition of spyware based on the Catalangate and other breaches:

> any product with digital elements that is specifically designed to exploit vulnerabilities in other products with digital elements and enables the covert surveillance of natural or legal persons by monitoring, extracting, collecting or analysing data from such products or from the natural or legal persons using such products, in particular by secretly recording conversations or otherwise using the microphone of a terminal device, filming natural persons, machines or their surroundings, copying messages, taking photographs, tracking browsing activities, tracking geolocation, collecting other sensor data or tracking activities across multiple terminals, without the natural or legal person concerned being specifically informed and having given their explicit specific consent.

They go on to describe the complexity of regulating and investigating spyware compared to wiretapping: "Contrary to conventional wiretapping, which only allows for the real-time monitoring of communications, spyware can provide full, retroactive access to files and messages created in the past, passwords, and metadata about past communications" (Veld, 2023).

During the incident, the US Department of Justice was able to observe the incident, as they were alerted to the breach by WhatsApp before they even patched it (Farrow, 2022).

3.3 Software Producer

WhatsApp and Facebook first detected the threat posed by Pegasus in May 2019 and quickly determined that it was an active exploit, affecting users as they were analyzing it. Initially, they decided not to block the attackers in order to get more time to analyze the vulnerability and its impact on users. This allowed them to trace the data, which ultimately led them to identify NSO, the creator of Pegasus, as the culprit. At this point, they were ready to update the software to patch the vulnerability, but before doing so, they alerted the Department of Justice. After implementing the update, they continued to monitor the behavior of the attackers, and also notified users that had been targeted (Farrow, 2022).

4 Responding, Recovering, and Remediating

Following the publicization of the data breaches in Catalunya, multiple widescale efforts by a variety of stakeholders began, including a broad campaign of disclosure by user advocacy groups, regulatory investigation and enforcement, international litigation, political posturing and public outcry.

4.1 Users

Targeted users may have first become aware of the breaches in 2019 when they were notified by WhatsApp of the vulnerability (Farrow, 2022). The incidents were then made public in an article in The Guardian in July 2020, as well as through the efforts of the Pegasus Project, and subsequently investigated by Citizen Lab, resulting in their 2022 report, which revealed the full extent of the breaches (Kirchgaessner & Jones, 2020; Scott-Railton, 2022; Amnesty International, 2022).

The immediate response to the breaches can clearly be seen as posturing: the targeted users made statements to journalists about the breaches and networked with activist organizations like Citizen Lab and Amnesty International. Those organizations also made public statements about Pegasus and related spyware.

Because many of the users were government officials, they were ardently involved in correction, compensation, and adherence behaviors at the national and international

levels. At the national level, they pursued their protections under Spanish laws regarding the use of spyware, and filed as many as nine complaints in court. In adhering to their rights under the regime, they sought an ex post investigation into the breaches through the independent Spanish Ombudsman's office, which oversees the use of surveillance technology by government authorities (Pujol, 2022). Additionally, in May 2025, two criminal complaints, leveraging different jurisdictions, against the Spanish CNI were announced. One by former regional president Artur Más and another by five plaintiffs, represented by an organization called Sentinel Alliance (Roussi, 2025). This litigation can be seen as correcting and/or compensating behavior, depending on the details of the complaint. At the European level, affected users and other civil society groups participated in the European Parliament's Pegasus Committee hearings and investigation in order to amend the regime to improve protections for potential victims of spyware (Veld, 2023).

4.2 Regulators

European regulators had shown interest in the Pegasus spyware prior to the Citizen Lab report, but within weeks of its release, official investigations were launched. Spanish regulators also responded to the incident, particularly in response to the public outcry and political action of the supranational European Union. Most of the responses seeking compensation or other remedies have been targeted at the perpetrators of the attack, not WhatsApp, which failed to prevent it.

4.2.1 European Regulators

Adhering to their mandate, the European regulatory regime responded to the data breaches of European citizens in Catalunya as a result of the Pegasus spyware. The European Parliament was alerted to the breaches, and as part of their enforcement authority, established a Committee of Inquiry to investigate, known as the PEGA Committee. The investigation included data breaches resulting from the use of Pegasus spyware in Catalunya and Spain as well as more than a dozen European countries (Veld, 2023).

There were multiple steps in the regulatory response process, including submitting legislation and eventually creating the PEGA Committee, which issued reports based on their investigation, culminating in a final one formally adopted by the European Parliament. Investigative artifacts like the PEGA Committee Study, *The impact of Pegasus on fundamental rights and democratic processes,* served as posturing

statements from the EU that highlighted the negative impact of spyware on democracy and individual rights (Sartor & Loreggia, 2022). The final PEGA Committee report was adopted by Parliament on May 8, 2023. Later, in June, the Parliament also made recommendations to the European Council and the Commission, as well as to Spain specifically (and other countries) on what further actions to take to correct the harms from the use of spyware (European Parliament, 15 June 2023).

The adopted PEGA Committee Report concluded that "Spain has an independent justice system with sufficient safeguards" but that there was room for improvement (Veld, 2023). Furthermore, the Parliament declared that "the regulatory framework in Spain is in line with the requirements set by the Treaties" but that "some reforms" are needed such as "real and meaningful legal remedy" for people who have been targeted by spyware (European Parliament, 15 June 2023). They called on Spain to undertake six corrective and enforcement actions, including investigation into the alleged cases of spyware use by the CNI, sharing judicial authorization materials with the victims, ensuring the victims have legal remedy for compensation, amending the legal framework of the CNI, and inviting Europol to contribute to the investigations (European Parliament, 15 June 2023). The stances by the Special Rapporteur and the European Parliament show the difference between complying with regulations and adhering to them: both documents describe Spain's regime as adequate, merely complying with the rules, but they also say that they could advance their protections of their citizens by adhering more closely to the spirit of the regime.

The PEGA Committee and subsequent Parliamentary texts called more broadly for a variety of actions to amend the regime governing the use of spyware, including legislation and action from the executive branch of the EU, the European Commission. The areas they specify for action include better implementation and enforcement of existing legislation, international cooperation to protect citizens, strengthening the capacity of telecom providers, bringing in Europol for more action, creating new entities to support research and development, and more (European Parliament, 15 June 2023).

Since the 2023 actions in Parliament, the main response behaviors have been plenary debates in Parliament and statements from MEPs, particularly as further news about the use of spyware in Europe is released (Perez, 2024; CDT, 2025; Jones, 2025). In early 2025, the rapporteur responsible for the PEGA Committee Report and Dutch MEP until 2024, Sophie in't Veld, made a public statement saying that legally-binding action needed to be taken by the EU to transform the regime that governs the use of spyware. The debates are simple compliance with legislative procedure, follow up discussions after the initial investigation. For example, key members of the European Commission and Council did not attend the 2024 plenary debate which had spyware and the PEGA

Committee on the agenda (Perez, 2024). The public statements can be seen as posturing to drive future action.

It is also worth noting that these responses have primarily been focused on amendments to the regime that governs “illegal” use of spyware. While Spain was largely found to be operating within the law, other Member States were excoriated for illegal or anti-democratic behaviors (Perez, 2024; CDT, 2025; Jones, 2025; European Parliament, 15 June 2023; Veld, 2023).

A countervailing voice came from MEP Jordi Cañas Pérez and an academic who published a report that called into question the validity of the data collected by The Citizen Lab and the PEGA Committee in Spain and Catalunya. In this report, they contribute to the Spanish authorities’ stance that the surveillance was conducted legally and with legitimate national security interests in mind, in full adherence with the law. They also state that much of the data about the potential “victims” of the surveillance was not able to be validated, and therefore could not be relied on as evidence of nefarious behavior (Olivas Osuna, 2023). This report, in particular the statement from the MEP included, serves to posture the regulatory response against the claims of the users that were surveilled, and in favor of the existing regime, encouraging no corrective behavior. Additionally, it shows a network formed with the MEP and an academic researcher to assist in that posturing behavior at the international level.

4.2.2 Spanish Regulators

Key responses from Spanish national authorities came from the CNI itself and the primary Ombudsperson’s office, known as the Defensor del Pueblo, an independent authority with investigative powers. According to a news article, the director of the CNI Paz Esteban, reported that they spied on “eighteen pro-independence leaders and people in their circles” with prior judicial authorization (Tomás, 2022). This leaves a difference of nearly 50 people compared to the report from Citizen Lab (including users that were targeted with software other than Pegasus). The Ombudsman, in compliance with their mandate, investigated and made a limited statement. In it, he declared that the CNI “acted in accordance with the Constitution and the law in the cases examined” (Pujol, 2022). In terms of correcting or enforcing, this could be considered no correction, maintaining the status quo of the regime and providing no compensation for the victims. The response does not even include mention of WhatsApp or Meta, the software producers. Together, these responses construct an *ex post* jurisdiction, that supports the *ex ante* operations, and the extant regime, that permitted the use of the spyware for national security interests.

In the judiciary system, new court cases have arisen in 2025, opening another door to correction of the regime, or compensation for the victims. As of September, one case filed by victims against the Civil Guard and the CNI, as well as the creators of the spyware, NSO Group, has progressed in a court in Barcelona. However, a previous case in the national courts was dismissed (Ara, 2025; Roussi, 2025). Spanish judicial authorities also attempted to investigate the spyware technology by sending a letter to Israeli authorities, where the NSO Group is headquartered, however, there was no response (Reuters, 2023).

In 2023, a report was released challenging the investigation by the Citizen Lab. This report, written by an academic researcher in cooperation with a Spanish MEP is an example of posturing in favor of the regime that governed the use of the spyware, and supporting the regulators' compliance behaviors. The report challenges the "methodological and ethical issues" of the Citizen Lab investigation, including the research design, field work, and findings. Among the 22 issues, the report claims that the claims of spying specifically by the lawyers representing the targeted users were used to try to upend their court cases, and that much of the monitoring was generally legal because many of the affected users were incarcerated. The report ties the Catalan independence movement with Russian involvement, which, by extension, implicates national security. The author included a statement from Spanish MEP Jordi Cañas Pérez that outlined his concerns which he also communicated to the PEGA Committee (Olivas Osuna, 2023).

4.2.3 Other International Regulators

There were some responses from international regulatory stakeholders following the publicization of Catalangate and similar incidents. The UN High Commissioner for Human Rights made a public statement calling for countries not to violate human rights, but also stated that surveillance measures can "be justified in narrowly defined circumstances, with a legitimate goal" (Bachelet, 2021).

In 2023, a joint statement by a network of 11 countries was issued stating their commitment to "counter the proliferation and misuse of commercial spyware" and an additional six countries added their names to it in 2024 (Herpig & Paulus, 2024; UK Foreign, Commonwealth & Development Office, 2024). Following this, in 2024, the UK and France began a new initiative calling for regime amendments governing commercial spyware, called the Pall Mall Process, but nothing beyond statements and creating networks has occurred (Herpig & Paulus, 2024).

4.3 Software Producer

WhatsApp took a variety of actions following the incident, primarily directed at the attackers. First, while remediating the vulnerability, WhatsApp tried to obfuscate the effects of the patch to try and collect data about the attackers, in hopes they didn't notice that the update affected their exploit. This could help the software team correct their security business processes to pre-emptively stop future attacks from attackers, and also provide data artifacts for other response, remediation, and recovery actions. Additionally, the WhatsApp security team adhered to disclosure requirements and notified users that had been targeted by the attack as well as the US Department of Justice (Farrow, 2022).

Next, WhatsApp/Meta filed a lawsuit against NSO Group in US federal court in October, 2019 barely 6 months after the vulnerability was discovered (Meta, 2025). This complaint was a good example of posturing, as it gave WhatsApp and Meta the ability to make public statements, begin constructing jurisdiction in a novel way, and protect their interests. The case would prove to be novel, as it became the first "victory against the development and use of illegal spyware" (Meta, 2025). This was also a compensation activity to get money from the attacker, and a correcting activity to try and prevent future attacks by setting legal precedence in future court cases. The US District Court of the District of Northern California subsequently ordered the NSO Group to stop using Pegasus to attack users of WhatsApp (Goodin, 2025).

In this court case, Amnesty International, a global human rights advocacy organization that also works on privacy issues, along with Access Now and other privacy advocacy organizations, joined the network of supporters for WhatsApp when they filed an amicus curiae brief that highlighted the human rights concerns with the Pegasus spyware (Amnesty, 2025; Access Now, Amicus Brief, 2020). As a note, it is rare that a user privacy advocacy group would side with a software producer like Meta, as they typically attempt to improve privacy protections for users by such large companies. While they have connected on issues like content moderation and freedom of speech issues, this collaboration between privacy advocates and big tech companies does seem to represent a more impactful interaction, according to an expert on the topic (Interview C004).

5 Outcomes

The outcomes for all stakeholders are still playing out five years after the breach was initially discovered. In courts, legislatures, and the media, the costs, transformations,

and remedies are still being assessed. However, thus far, the regime has been preserved.

5.1 Users

The targeted users in this case are still seeking meaningful remedies. The Spanish authorities, specifically the Ombudsman, ruled that the data breach was lawful. This has meant that, thus far, there has been no compensation and no transformations to their protections, nor direct prevention from future harms. While there have been lawsuits filed, some have been dismissed and some are still in progress, but no compensation or transformations have come from these cases yet either. However, the broad public outcry about the use of spyware in Spain and elsewhere, especially in Europe, could lead to some transformations as advocates work together with targeted users to shame prospective perpetrators, specifically government actors.

The costs for these users could include things like a loss of privacy, impacts on their careers, or the need to adopt new processes to protect themselves. More broadly, it could also have a chilling effect on their involvement in civil and democratic action.

The day following the formal adoption of the PEGA Committee Report, a user advocacy organization called European Digital Rights (EDRi) plainly stated their opinion on the outcome in a report entitled “PEGA Committee does not go all the way on spyware regulation” (EDRi, 2023). The EDRi response is clearly critical of the result and actively seeks to transform the regulatory regime to improve user protections. First, the result is that there is a “conditional moratorium” on the use of spyware by Member States, requiring them to investigate and resolve spyware abuse cases, align national legal frameworks with international law standards, involve Europol in investigations, and repeal certain export licenses. EDRi claims that there would not be any enforcement action based on these rules, but that some of the concepts could lay the groundwork for future legislation (EDRi, 2023). This critique was mirrored by another user advocacy organization, Amnesty International in a brief public statement the same day (Amnesty International, 2023).

5.2 Software Producers

Meta and WhatsApp have been aggressive in their response, recovery, and remediation activities. The results have been largely favorable for them. In 2025, their lawsuit against NSO Group netted them over $4 million in compensation in US courts, and a permanent injunction on the use of Pegasus against WhatsApp users (Goodin, 2025). The decision highlighted that NSO Group violated the terms of service they would have

agreed to when accessing WhatsApp (Page, 2024). This outcome can be seen as a transformation in favor of other software producers as it sets precedent to protect them from future abuses of their technology when vulnerabilities are exploited for spyware and profit by third parties.

This case also brought privacy advocates into a cooperative network with software producers to fight a shared threat, which is not typical, and could have a lasting impact on the way these two stakeholder groups collaborate (Interview C004).

5.3 Regulators

In this case the regulators in Spain are also essentially the enforcers of the regime that enabled the attack to occur, making it a confusing situation to describe. In the European Parliament, regulators are limited in their ability to influence the national politics of Member States. In the US, courts awarded WhatsApp with damages, setting a new precedent for future cases.

5.3.1 Spanish Regulators

As a result of their enforcement actions, namely testifying before the legislature and investigating through the ombudsman's office, Spanish regulatory authorities predominantly preserved their power to use spyware in the future. They found that their actions did not violate any rules and therefore took no action. There was no compensation or remedy for the affected individuals, although only eighteen cases were investigated fully (Pujol, 2022; Tomás, 2022).

However, the courts have decided to hear at least one of the lawsuits against the entities that used the Pegasus spyware as of September 2025 (Ara, 2025).

5.3.2 European Regulators

The adopted PEGA Committee Report and the Recommendation of 15 June 2023 are not legally binding remedies, but they are both enforcement actions in their own right, documenting facts and proposals in the legal record.

Amnesty International and EDRi both responded to the PEGA Committee Report by saying that recommendations did not go far enough (Amnesty International, 2023; EDRi, 2023). In general, these responses describe the transformation in the regulations surrounding the use of spyware in Europe due to this event as modest, and that they are largely a preservation of the existing approach, including weakly enforced rules and strong deference to Member States in their handling of national security matters. On the other side, the report that criticized the Citizen Lab findings, which includes a signed

statement from a Spanish MEP, seems to support preservation of the regime, declaring that the surveillance was legitimate (Olivas Osuna, 2023).

There has only been limited action dealing directly with the investigations that concluded in 2023. There have been debates in Parliament and public statements from MEPs, which are not seen to be productive (see Section 4.2.1) (Perez, 2024; CDT, 2025; Jones, 2025; Veld, 2025). However, there have been some attempts at legislative changes that have impacts on the legal uses of spyware.

However, two regulations in particular have had an impact on the regime, to prevent future spyware threats, although spyware is not their only focus: the Cyber Resilience Act (CRA) and Media Freedom Act (EMFA), both passed in 2024. The CRA sets cybersecurity requirements for hardware and software available in the EU to help ensure they are more secure, including from threats such as spyware (European-cyber-resilience-act.com, accessed 14 October 2025). The EMFA adds protections for the media and journalists, including prohibiting the use of spyware against them, and requiring Member States to designate an independent authority to handle complaints. Once, again, however, there are limitations for national security and "serious crimes" such as terrorism (Media-freedom-act.com, accessed, 14 October 2025).

5.3.3 Other International Regulators

In 2024, France and the United Kingdom proposed a new regime for "Tackling the Proliferation and Irresponsible Use of Commercial Cyber Intrusion Capabilities" called the Pall Mall Process. The initiative focuses primarily on commercially available cybersurveillance software like Pegasus. It boasts a multistakeholder network including multiple national governments, industry, and civil society (Pall Mall Process, 6 February 2024).

However, according to a report from The Lawfare Institute in cooperation with the Brookings Institution, the process may be limited in its impact because it does not address the role that governments play in the proliferation and irresponsible use of spyware. They state that "governments are fueling the commercial cyber intrusion capabilities ecosystem with millions of taxpayer dollars every year" effectively driving the spyware software economy. Additionally, they argue that the Pall Mall Process does not directly address the capacity of governments to create their own spyware (Herpig & Paulus, 2024). A follow up conference was held in April 2025, but it does not appear that any transformations have resulted thus far.

In US courts, the Meta lawsuit against NSO Group took six years, but the end result was a landmark court decision that ordered NSO Group to “permanently cease targeting WhatsApp users” (Goodin, 2025).

6 Aftermath

As a result of the far-reaching impacts for all stakeholders in terms of regulations, operations, and threats in this case, I would argue that there has been a *systemic disruption* achieved based on Lucas Kello’s theories of virtual weapons and international order (2017). Kello describes systemic disruption as the lowest of three levels of international revolution arising from cyber threats. While there has been a major disturbance that “disrupts the conduct of units,” the overarching result is still continuity, as opposed to the other two levels. Among states, systemic disruption “affects the dealings of self-interested states pursuing imperatives of survival and welfare” and impacts social “rules, norms, and principals” between non-state actors. He provides four test principles for systemic disruption: technological volatility, the signature of power (attribution), offense-defense balance, and the depth of power (Kello, 2017).

The use of Pegasus as a weapon by governments against citizens has unleashed a multistakeholder effort for new regulatory controls, and better security. Additionally, there have been new relationships formed between civil society actors to challenge the use of such spyware. These multistakeholder responses meet the tests provided, and have crucial impacts on the regulatory, threat, and operational environments for each stakeholder group, ultimately altering the broad context for future cases.

Technological Volatility

Spyware technology has been regarded as a threat to human rights and potentially destabilizing to democracies. Regulators have started to mitigate this volatility going forward by improving security measures by mandate. For example, the Cyber Resilience Act (CRA) requires software producers (among others) to harden cybersecurity measures to protect their users against spyware attacks. This is achieved through training and certifications for cybersecurity professionals, improving the threat environment by increasing familiarity with vulnerabilities, and supporting regulatory compliance. (European-cyber-resilience-act.com, accessed 14 October 2025).

Furthermore, there has been public outcry by advocacy organizations like EDRi and Amnesty International to significantly restrict the ability of governments to purchase spyware technology (EDRi, 2023; Amnesty, 2023). While no such rules have been

passed, the public attention provides an additional impediment to future deployments. This improves the threat environment for users who have more resources to become familiar with threats, and improve their operations when using potentially vulnerable software.

Signature of Power

When an attack cannot be attributed to an attacker, responses are limited. And when the attribution is achieved, coordinating a response can be difficult and time consuming.

As a result of the CatalanGate case, European investigative authorities, journalists, and privacy advocates jointly disrupted the challenges with the attribution of cyber attacks, although it did take time. Most obviously, the forensic identification of the Pegasus software itself and its creator, NSO Group, was a substantial finding. From there, investigators were able to trace the government customers that purchased the software. In the case of CatalanGate, the Spanish government eventually admitted to conducting a portion of the breaches that occurred.

Additionally, the software producer, WhatsApp, definitively attributed and successfully challenged NSO Group in US court (Page, 2024), setting new precedent for future spyware cases in an additional jurisdiction.

Through such advances in attribution, the regulatory environment for all stakeholders is impacted, as it allows each actor to prepare for spyware threats, as well as pursue remedies should they be attacked.

Offense-Defense Balance

Cyber attacks typically favor the attacker, for a variety of reasons (Kello, 2017). However, the synergy of these multistakeholder responses in this case have confounded the offensive advantage of cyber attackers by trying to improve defenses from potential victims, and providing new methods of fighting back against future threats, across the whole class of spyware virtual weapons. The combination of improved security regulations and public remonstration affects the “dominance, usefulness, and perishability” (Kello, 2017) of spyware threats.

While there have been, and continue to be, examples of legislation, journalism, lawsuits, and enforcement against cyber threats, this case shows a combination of these efforts combining in a way that had novel results. The PEGA Committee, later legislative efforts, US court cases, and investigative journalism all came together to better balance defensive capabilities against offensive spyware users. Going forward, the operational environment of software producers, regulators, and users is disrupted because they now have experience collaborating on preventative measures, and new tools, to meet

each of their needs. This has been achieved through regulatory changes, public outcry, and enforcement, which brought these stakeholders together in a way that had not seen before.

Depth of Power

The Pegasus attacks occurred in multiple jurisdictions and regimes, presenting an immense challenge to coordinating a response. But, the result of this case has been a novel collaborative approach to mitigating them going forward. Once the attacks had been identified, representatives from each group of stakeholders were able to collaborate over the course of months and years to formulate a response that will help future confrontations with spyware threats. Where a single private actor, or a local government would be severely limited in their ability to form an effective response that could protect themselves from future attacks, the systematic multistakeholder collaboration in this case helped improve the “strategic depth” of defenses against spyware attacks. For example the amicus brief submitted by Amnesty International and Access Now in support of WhatsApp’s lawsuit, demonstrates a positive method for users and software producers to work together with court regulators in the future, improving the operational and regulatory environments for each stakeholder.